

Weekly Learning Reflection

Machine Learning — Week 1

1. Satisfaction with My Progress

This week was hard, but I gave my best. I covered all 12 topics on my curriculum and kept going even when concepts felt difficult. I am not fully satisfied yet because some areas still need more practice, but I am proud that I did not skip anything.

2. ML Topics I Covered This Week

- Artificial Intelligence and Machine Learning — what they are, how they differ, and real-world applications
- Machine Learning Lifecycle — data collection, preparation, training, evaluation, deployment, and improvement
- Role of Data in ML — structured vs unstructured data, features and labels, training/testing splits, data quality
- Python Basics — variables, data types, operators, control statements, input and output
- Logical Problem Practice — prime numbers, unique numbers, and basic coding challenges
- Basic Statistics — mean, median, mode, variance, standard deviation, range, and quartiles
- Basic Mathematics — vectors, matrices, and basic probability (independent and dependent events)
- Modern AI Systems — traditional ML, deep learning, generative AI, LLMs, and agentic AI
- Development Environment Setup — Python, pip, virtual environments, Jupyter Notebook, VS Code
- Problem Solving for ML — thinking in inputs/outputs, breaking problems into steps, experimentation mindset
- AI-Assisted Learning — using AI for explanations and research, risks of blind trust, validation practices
- Hands-on Tasks — set up environment, explored datasets, documented ML lifecycle, compared traditional vs ML systems

3. How These Topics Contributed to My Growth

Studying these topics gave me a clear picture of how ML works end-to-end. I can now explain what AI and ML are, set up a working development environment, write basic Python programs, and understand the math(NOT AT ALL) ML models.

4. Mistakes I Want to Avoid Going Forward

- I need to understand concepts deeply, not just read them quickly
- I should practise writing code from memory, not just follow examples
- I must not rush — mathematics need more time and repetition
- I will validate any AI-generated explanation instead of accepting it directly

5. Strategies I Am Adopting Differently

- I will summarise each topic in my own words after studying it
- I will use short focused study sessions instead of long tiring ones
- I will practise one small coding problem every day to build my Python skills

Overall

This Week 1 learning reflection captures a developing machine learning student's journey through foundational concepts with notable discipline and self-awareness. Despite finding the material challenging, the author persisted through all twelve planned topics, ranging from core AI/ML distinctions and the machine learning lifecycle to practical skills like Python programming, development environment setup, and statistical foundations including mean, variance, and probability. The reflection demonstrates genuine engagement rather than surface-level coverage, touching on data fundamentals, modern AI systems like LLMs and generative AI, and hands-on exercises such as dataset exploration and lifecycle documentation.

What stands out most is the honest self-assessment: the writer acknowledges incomplete mastery rather than overstating confidence, recognizing that statistics and mathematics need deeper repetition rather than quick reading. The reflection closes with a thoughtful, actionable improvement plan, including daily coding practice, prioritizing weak areas, validating AI-generated explanations instead of blindly trusting them, and using shorter, focused study sessions over long exhausting ones.

Overall, this entry reflects a learner who is building not just technical knowledge but also metacognitive habits, self-discipline, and a sustainable study approach essential for long-term growth in machine learning.